Lesson 4-3 · Period 3

Rational Expression Modeling + Assessment-Critical Simplify

Klimsara-adapted · Student-centered · No Blooket

Do Now — Identity and Domain Algebra 2 · Lesson 4-3 · Period 3

DOK 2 | 5 min | *bridge from P2*

Do Now · Identity check (4-3-savvas-q11) **Reason**

Explain why

$$\frac{4x^2 - 7}{4x^2 - 7} = 1$$

is a valid identity — and state the values of x that must be restricted from the domain.

Launch — Example 2 + Example 6 (back-to-back)

DOK 2 | 12 min | *teacher models Ex 2 + Ex 6*

DOMAIN RULES (mark these — assessment reference)

1. Domain = all reals EXCEPT zeros of **every** denominator in every step.
2. Write the simplified form AND the restriction list together.
3. Canceled factors still exclude their zeros.

Explore — TPS: 6 items in order (35 min)

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DOK 2 | 35 min | *student work*

Abstract forms first, then modeling, then assessment rehearsal. Never cut #19.

Try It 6 · SA/V comparison (two prisms) Which container has the smaller surface area-to-volume ratio?

Practice #14 · Domain restriction reasoning Why are restrictions needed to show $\dfrac{-6x^2+21x}{3x}$ and $-2x+7$ are equivalent?

Practice #22 · Simplify + identify domain Simplif

Share / Summary — Assessment-Critical Reveal

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DOK 2 | **5 min** | *DOK-2 reveal + Topic 4 assessment preview*

Sentence frame for #19 (pair presents, class signals) "The denominator $x^3 + 3x^2 - 10x$ factors as __________.
The zeros are ____, ____, ____, so $x \neq$ __________.
A cubic denominator may have up to __ restrictions."

Topic 4 8-Q Assessment — Q#2 is this exact content Practice #19 directly rehearses Topic 4

Exit Ticket — Pre-Assessment Confidence Check

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DOK 1-2 | 3 min | *not graded*

Complete on your exit ticket

1. A rational model needs factoring because ________________________.
2. My simplified ratio is _____________, valid when $x \neq$ ________.
3. In context this means ________________________.
4. One biggest thing I learned today: _____________________.